Big Data Analytic for Estimation of Origin-Destination Matrix in Bus Rapid Transit System

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***Abstract*—In the field of transportation, the origin-destination matrix is one of the main and important components, especially in analyzing, planning, and managing a public transport network. The traditional survey can be used to determine passengers travel patterns and generate origin-destination matrix, but it is inefficient and cost a lot of resources. In the recent years, various methods have been studied to estimate origin-destination matrix to reduce costs and increase the accuracy of passengers flow. Many of them take advantages of big data technology to gather passengers travel information, mostly using smart card data. In this paper, we perform origin-destination matrix estimation using information from the smart card that was collected from automatic fare collection systems in Jakarta's Bus Rapid Transit. There are approximately 160 million records from 20 months of transactions between June 2014 and January 2016. This study utilized trip chaining algorithm that generates 610 daily OD matrices, 87 weekly OD matrices and 20 monthly OD matrices. The analysis is performed at a station and a line level, with an addition of passenger behavioral pattern.**

Keywords— transportation; bus rapid transit; origin-destination matrices; smart card; automatic fare collection systems; big data

I. INTRODUCTION

Public transport is an important aspect of major cities around the world. Various public transport systems implemented to accommodate citizens' needs of travel, and one of those public transport systems is Bus Rapid Transit (BRT). Jakarta, as the most populous city in Indonesia, has implemented BRT as one of the main public transport for its citizens. Jakarta's BRT is the first of such system in Southeast Asia and has the largest and longest network of dedicated BRT lanes in the world, which approximately is about 210 km [3].

Recent statistics show that Jakarta's BRT has been transporting more than 111 million passengers a year [4]. The growth of passengers pushes authorities to improve operational of the system and their services, so that passenger's satisfaction increases. Thus, more people are expected to use public transport, which could have an impact in reducing the city's level of congestion.

In planning and management of public transport system, the information about how passengers travel becomes important. Origin-Destination (OD) matrix is often used by transportation engineers to answer this question. OD matrix is a matrix that shows the total numbers of trips between origin and destination in the transportation network. It provides the information where the trips begin and where they end [1].

In the recent years, various methods have been used to generate OD matrix in the public transport network, mostly using information from smart card data which is collected by automatic fare collection systems. Most of the researches use trip chaining method to infer the boarding and alighting station and estimate the OD matrix by using two assumptions, the boarding station of their next trip is the alighting station of their previous trip, and the alighting station of their last trip of the day is the boarding station of their first trip of the day [5][6][7].

Estimating OD matrix from huge numbers of data is a

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challenging task, and the right method is required to handle and process the data. Thus, big data approach becomes important and used in this paper. Apache Spark and Hadoop Distributed File System (HDFS) were used in this study. Furthermore, different visualization techniques are needed to make the data easier to understand, and help us analyzing passengers behavior.

This study focuses on estimating OD matrix using information from smart card data, which is collected from automatic fare collection systems. In section II, data description is elaborated. Section III explains how to generate OD matrix, starting from preprocessing, data exploration, and OD matrix estimation. Section IV provides data exploration results. In section V, research conclusion is presented.

II. DATA DESCRIPTION

This research uses Jakarta's public transport as a case study. Jakarta is the capital city of Indonesia, located on the northwest coast of Java island. Jakarta's covering area is 661.52 km², divided into 6 cities and 34 districts, with an estimated population of over 9 million people. According to Jabodetabek Transportation Agency Business in 2017, public transport users in Greater Jakarta reached 40.5 million transactions per day.

In this research, Jakarta's BRT transactions data, which is obtained by using a smart card and automatic fare collection system, is used. There are approximately 160 million records from 20 months of transactions between June 2014 and January 2016. This data is collected from 296 gates in 224 stations of the Jakarta's BRT system, which are grouped in 12 lines.

III. TRIP ESTIMATION

Three main phases are used to generate OD matrix: preprocessing, data exploration, and OD matrix estimation. Some analysis for data transaction in December 2015 will be given. Preprocessing section explains how to prepare the data before OD matrix estimation. In data exploration section, the tendency of a transaction conducted by passengers is explored. OD estimation section explains the algorithm and the process of generating OD matrix.

A. Preprocessing

BRT transaction data is obtained from June 2014 to January 2016. Originally, the data is in the form of six MySQL tables, with the total size of around 65 GB. The tables were converted into comma separated value and then stored it to HDFS. Apache Spark on top of Hadoop environment is used to handle big data. After conversion, join operation between files were performed and certain features to be used to generate OD matrix were selected. The features are card_id, datetime_transaction, and boarding_station_information. Afterwards, the data is saved as parquet file format to make it faster to be processed later and to significantly reduce the size.

The result is the clean data that ready to be processed further. The data is comma separated value file which have been grouped by card_id and sorted by datetime_transaction to make it easier in estimating OD matrix.

B. Data Exploration

In December 2015, the total number of transaction was 8.3 million, with the daily average is 270,000 transactions. Analysis of transactions pattern every single day is performed. Afterwards, an average number of transactions per hour in December 2015 is calculated, as illustrated in Fig. 1.

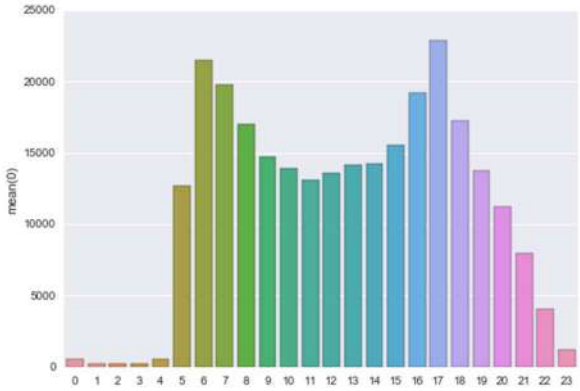

Fig. 1. Analysis of BRT transactions

Fig. 1 shows that the highest transaction occurs at 17.00 until 18.00 and 06.00 until 07.00, which is the time when most people are going home and going to work, respectively. The number of transaction decreases at 07.00 until 10.00, from around 20,000 to around 14,000 transactions, and there is not a lot of changes in the number of transaction at 10.00 until 15.00. Subsequently, the transaction at 15.00 until 17.00 increases significantly, from around 15,500 to around 23,000 transactions. From 18.00 to 23.00, the number of transaction decreases significantly. It can be seen that the quietest hour occurs between 23.00 and 05.00, with only 300 - 500 transactions per hour.

This study also analyzed daily transactions in December 2015, and calculated the average of transactions every day of the week. The result is shown in Fig. 2.

Fig. 2. The average number of transactions for each day

Fig. 2 shows that the highest transaction occurs on Monday with the total transactions is around 297,000, followed by Tuesday with around 296,000 transactions. On Wednesday, Thursday, and Friday, the number of transaction decreases to around 278,000 transactions per day. The lowest

transaction occurs on Saturday and Sunday, with only around 233,000 transactions per day.

The third analysis is about the behavior of individual transaction. The result is derived from the descriptive analysis that was performed by calculating the total number of transactions per individual passenger / card holder, as can be seen in Fig. 3.

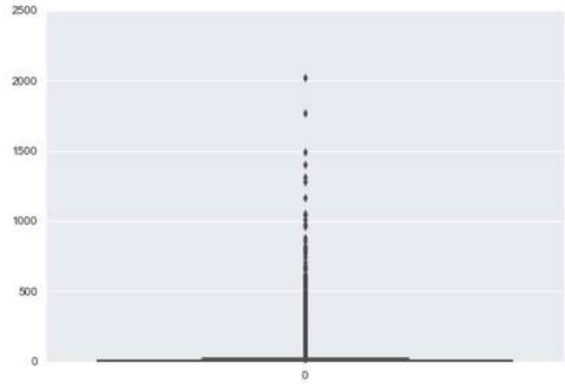

Fig. 3. Total transactions by each individual passenger

The result shows that there are 1,037,001 unique card_id that did transactions in December 2015. The average transactions conducted by each card_id is 8 with median 4 and standard deviation 12. The difference between the average and the median occurs because there are 93,021 card_id which have the number of transactions beyond the reasonable limit, and the highest number of the transaction conducted by one card_id in December 2015 is 2022 transactions.

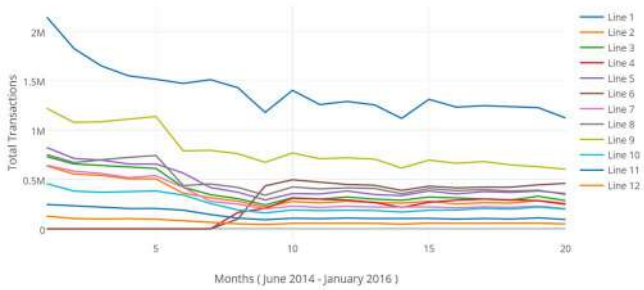

Fig. 4. Total transactions in 20-month period by each line

Fig.4. shows the total number of transactions in each line during the 20-month period, from June 2014 to January 2016. It is seen that there is decreasing in total transactions in every line in the first eight months. As shown in Fig. 4., the fourth and sixth line do not record the transactions in the first 6 months because the automatic fare collection system is newly implemented there in December 2014, or on the seventh month in the graph. Overall, there is no significant difference in the total amount of transactions in the past 10 months.

C. OD Matrix

Trip matrix or OD matrix is a two-dimensional matrix which are the column and row represent a zone in the study area.

Zones	1	2	...	j	...	n	O_i
1	T_{11}	T_{12}	...	T_{1j}	...	T_{1n}	O_1
2	T_{21}	T_{22}	...	T_{2j}	...	T_{2n}	O_2
⋮	...	...	...	...	...	...	⋮
	T_{i1}	T_{i2}	...	T_{ij}	...	T_{in}	O_i
⋮	...	...	...	...	...	...	⋮
n	T_{ni}	T_{n2}	...	T_{nj}	...	T_{nn}	O_n
D_j	D_1	D_2	...	D_j	...	D_n	

Fig. 5. OD matrix illustration

The cells of each row i contain the trips originating from zone i which have destinations j in the corresponding columns. T_{ij} shows the number of trips between origin i and destination j. O_i is the total number of trips originating from zone D_j and i is the total number of trips attracted to zone j [2].

The algorithm that is used for determining the OD matrix is shown in the Fig. 6 as follow:

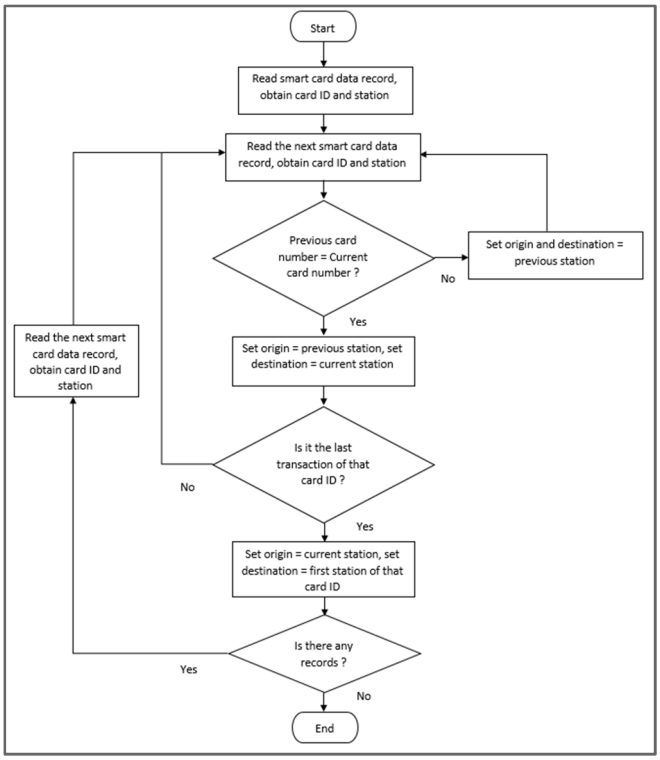

Fig. 6. Flowchart for determining OD matrix

The purpose of the algorithm is to detect movement between station for each individual passenger. In this algorithm, if a passenger only performs one transaction in a day, the origin and destination are set at the same station. If a passenger performs more than one transaction in a day, the station from the first transaction is counted as the origin and also the last destination of that day. Station in the next transaction has two functions, as a destination for a previous

trip and as the origin for the next trip. This will be repeated until it reaches the last transaction of that passenger in that day. Finally, the last trip's origin is the station in the last transaction, and the last trip's destination is the station in the first transaction performed by a passenger in that day. Because Jakarta's BRT system only records the passengers boarding station, and does not record passengers alighting stations, estimation is performed using assumption based on trip chaining method to generate OD matrix.

In this research, three types of OD matrices were created. Those are OD matrix based on transactions in a day, OD matrix based on transactions during a week and OD matrix based on transactions in a month. Weekly and monthly OD matrices are created because there are the cross-days transaction possibility and daily OD matrix is not capable of capturing it. For instance, when passengers depart close to midnight and arrive at the alighting station after 12 pm, which counted as the next day, instead of records it as one trip, daily OD matrix will calculate it as two different trips in two different days.

IV. DISCUSSION

Using the algorithm, this study generated 610 daily OD matrices, 87 weekly OD matrices and 20 monthly OD matrices. Moreover, additional analysis was performed onto it, for instance here, is the analysis of transactions that occurred in December 2015.

The first analysis was performed at station level. The station is a temporary bus stop where passengers can board and alight. By using the station as a unit to determine OD matrix, passengers movement analysis will be easier to understand. We performed analysis to determine the distribution of departures from each station and then visualized it, as can be seen in the Fig.7 boxplot.

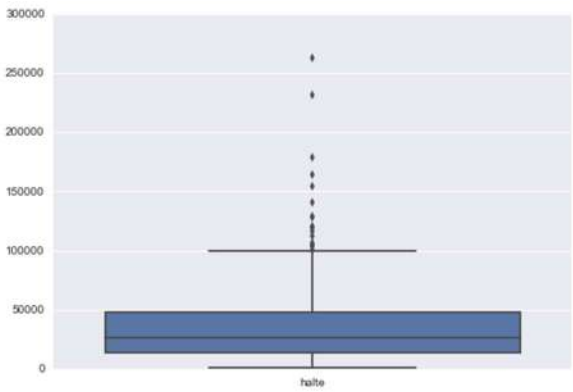

Fig. 7. Distribution of departures by each station

The transactions were obtained from 224 stations, with the average number of departures made by each station is 38,007, with the median of 26,267, and standard deviation of 38,462. Station with the highest number of departures is Blok M station, which records 262,589 departures in a month. Fig. 7. above shows that certain stations are the outlier because their number of departures are more than 1.5 * interquartile of upper quartiles.

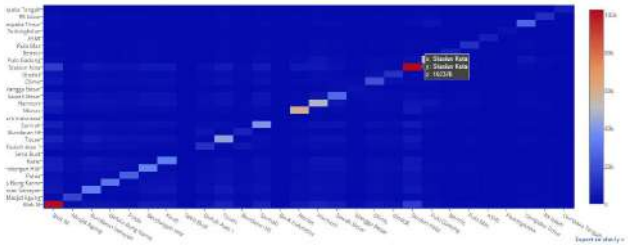

Fig. 8. OD matrix heat map visualization at station level with diagonal values

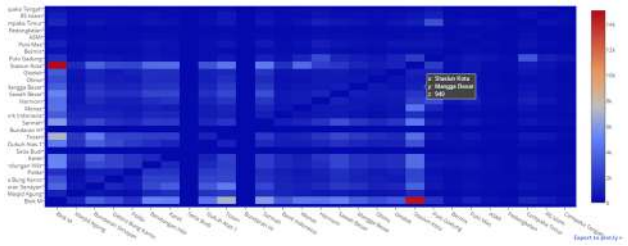

Fig. 9. OD matrix heat map visualization at station level without diagonal values

To make the existing OD matrix easier to understand, it is presented as a heat map visualization. As can be seen in Fig. 8, trips between the same station are dominating. It means there are a lot of trips that their alighting stations are unknown.

In Fig. 9, the figure do not include the diagonal values, it shows that the trips are not unevenly distributed because there is a difference in the variation of the color from the cells. It can be seen that there are two cells with the most contrast color, which show the higher number of trips. Red, white, and blue indicate a high, medium, and a low number of trips respectively. In the example above, the y-axis is the name of origin stations while x-axis is the name of the destination stations and z represents the number of trips. For instance, at y = Mangga Besar, x = Stasiun Kota and z = 940 indicate there are 940 trips from Mangga Besar station towards Stasiun Kota station.

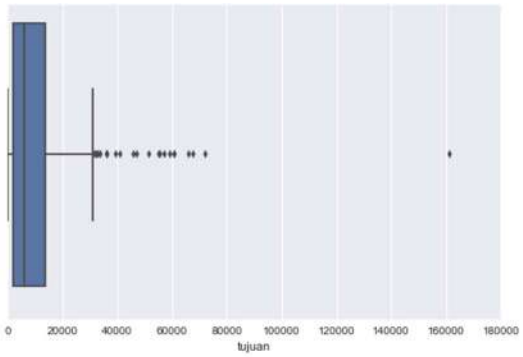

Fig. 10. Distribution of arrivals by each station

To obtain the total number of arrivals, all number of trips is added up, except those in the diagonal side of OD matrix because it indicates the trips that the alighting stations are unknown. We performed a descriptive analysis of statistics regarding the results. It shows that the average number of

arrivals in each station is 11,997, with a median of 6,090, and a standard deviation of 17,479. The highest number of arrivals in one station in a month is 161,460, and the result shows that the number of arrivals in each region is quite varied.

There is a big difference in values between the total number of departures and arrivals, even though it should not happen in an ideal circumstance. This occurs due to the anomalies in the data.

The second analysis was performed at line level. It will not provide the information about how people move from point A to point B, but how Jakarta's BRT line design could meet passengers needs. For instance, whether the passengers need to change the line to get to their destination.

In this data analysis all of the 12 Jakarta's BRT lines are used. The average number of departures of each line is 694,923 with a median of 685,614, and a standard deviation of 426,421. During the period of December 2015, the highest and lowest number of departures from one line is 1,807,701 and 134,559 respectively. The distribution of these values can be seen in Fig.11 which shows that there are a few outliers in both of upper number and lower number of departures.

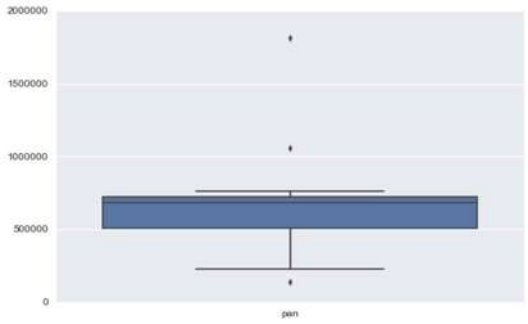

Fig. 11. Distribution of departures by each line

Two visualizations of OD matrices at line level were made. The first one is OD matrix with diagonal values visualization (Fig.12) and the second is OD matrix without diagonal values visualization (Fig.13).

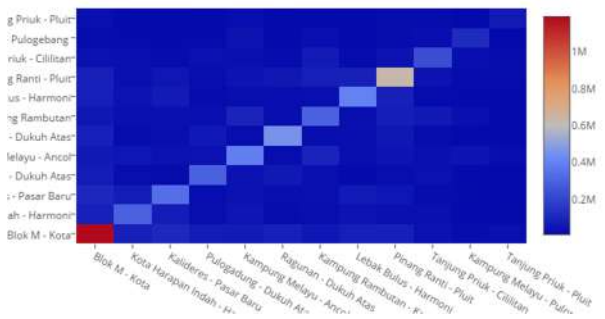

Fig. 12. OD matrix heat map visualization at line level with diagonal values

In OD matrix visualization at line level that still includes the diagonal values, it can be seen the diagonal cells look brighter than the non-diagonal cells. This indicates the origin-destination station pairs are mostly still in the same line, because a number of trips performed by passengers in the same line is higher than the trips of passengers who need to change to the other line to get to their destination.

In visualization that is eliminating the diagonal section (Fig.13), it can be seen more color variations of the OD matrix. This shows us how passengers were doing trips by changing the line to get to their destination.

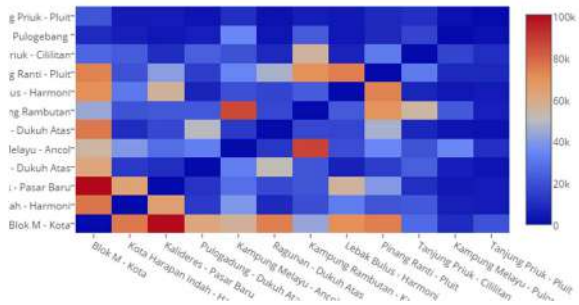

Fig. 13. OD matrix heat map visualization at line level without diagonal values

From the results of the previous analysis in OD matrix at station level, it was found several anomalies in the data that led to the low number of arrivals and the high number of trips which the origin and destination station are the same. It means that we are unable to infer all of the alighting stations precisely. Some analysis of the causes of the anomalies are as follows:

- There are passengers who do not have their own smart card and they use a single trip card, a smart card provided by the operator in each station. In fact, one single trip card is used by all of the passengers who do not have a smart card, so one card_id is able to record a lot of transactions, which does not reflect the real situation.
- There is a high number of passengers who only travel once during the period of December 2015, which is around 194,000 passengers.
- There are many transactions conducted by one passenger in a day but they all have the same boarding stations. This can be caused by two things:
  - It is possible that a passenger returns to the previous station without using BRT, and then departs again in the same station as before. The system then records that passenger departs from the same station in sequences at different times in a day. This case occurred as many as 1,439,377 times during December 2015.
  - Passengers depart in a group but using only one smart card for boarding, this can be identified by looking at a passenger who performed two times or more transactions in a very short lapse of time. We found there are 1,442,692 group of transactions for this case.

V. CONCLUSION

OD matrix can be used effectively for understanding passengers travel pattern. This study has succeeded to map passengers travel pattern into OD matrix using information from smart card data in Jakarta's BRT system. However,

36.5% transactions the alighting stations still difficult to be inferred because of passenger's certain behavior. The deeper analysis is also needed to give us more understanding about how people travel, for instances how often people use public transport in a group, or in what day people are more likely to use public transport rather than a private vehicle, and so on.

Future researches can focus on how to deal with passengers characteristic and behavior to infer the right alighting stations, and to increase the accuracy of OD matrix.

VI. ACKNOWLEDGMENT

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